Experiment (2)

Bleeding of mice, and Serum and Plasma Preparation

IMMUNOLOGY & SEROLOGY LABORATORY
(BT336)

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Introduction

Bleeding of experimental animals is needed to prepare normal serum and immunized serum.

Successful bleeding requires

Care.
Patience.
Practice.

What is the Blood?

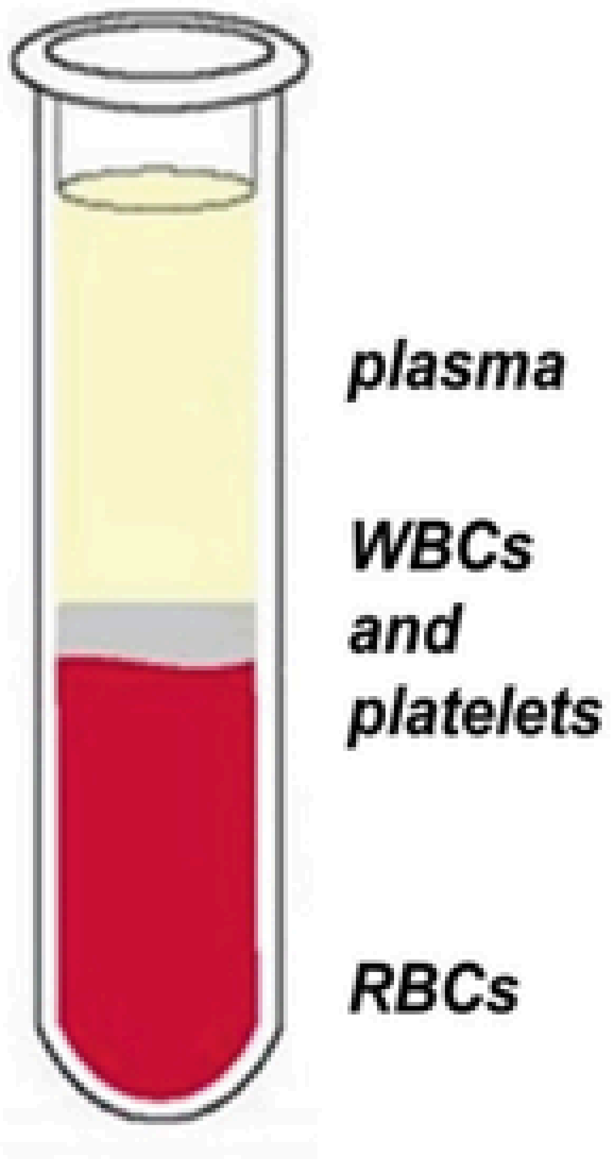
- The whole blood is composed of cellular elements and plasma.

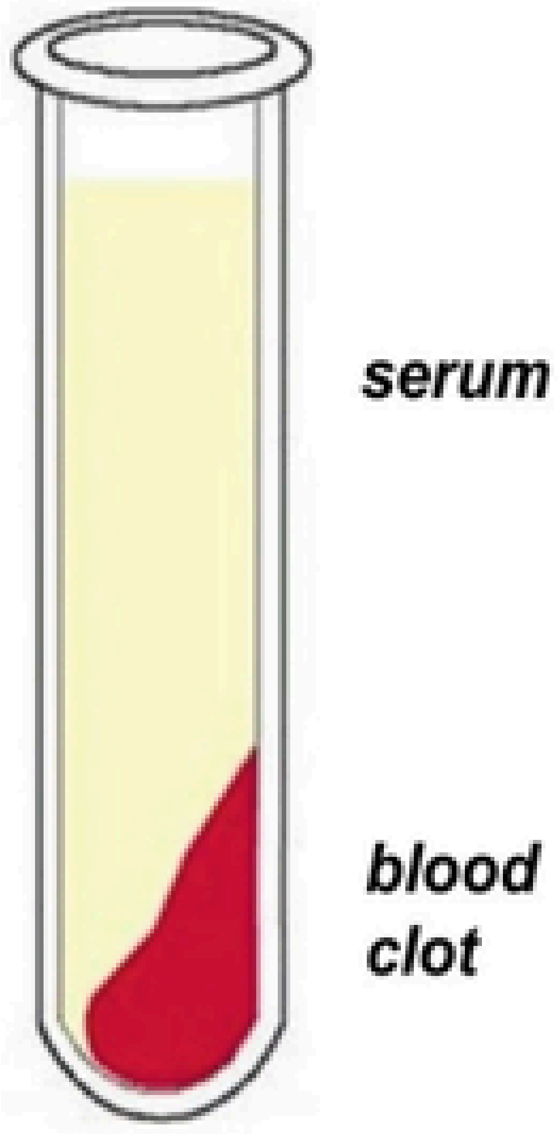
- Upon clotting of fresh blood, the soluble fibrinogen in the plasma is converted into insoluble fibrin and the remaining fluid is the serum which is a clear yellowish fluid.

- Although fibrinogen forms only 3% of the plasma it interferes with most types of serological tests.

What is the Difference between Serum and Plasma?

- **Serum** is the liquid fraction of whole blood that is collected after the blood is **allowed to clot.**
  - The clot is removed by centrifugation and the resulting supernatant, designated serum, is carefully removed using a Pasteur pipette.
- **Plasma** is produced when whole blood is collected in tubes that are **treated with an anticoagulant.**
  - The blood **does not clot** in the plasma tube.
  - The cells are removed by centrifugation. The supernatant, designated plasma is carefully removed from the cell pellet using a Pasteur pipette.





PLASMA	SERUM
anti-coagulants are needed for purification	anti-coagulants are not needed
it can be prepared as soon as it has been mixed thoroughly	30 minutes delay for a clot formation
fibrinogen is present	fibrinogen is absent
platelets and cells (WBCs) can contaminate the liquid fraction	cleaner sample, depleted of cells and cell remnants, but latent clotting can lead to fibrin formation
composition of ions is representative of the circulating blood	clot retraction elevates potassium level relative to its plasma value
considered less stable (especially during longer storage)	considered more stable – the gold standard for biobanking

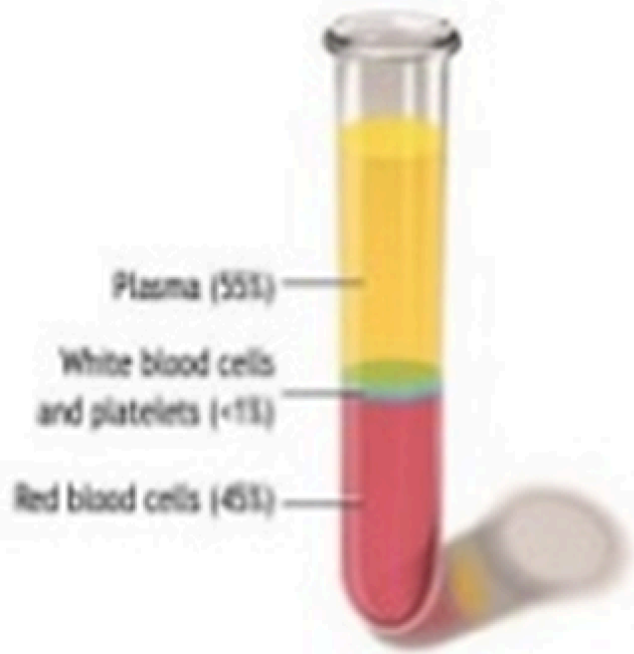
2. Plasma vs. serum

•**Plasma** is the liquid, cell-free part of blood, that has been **treated with anti-coagulants**.

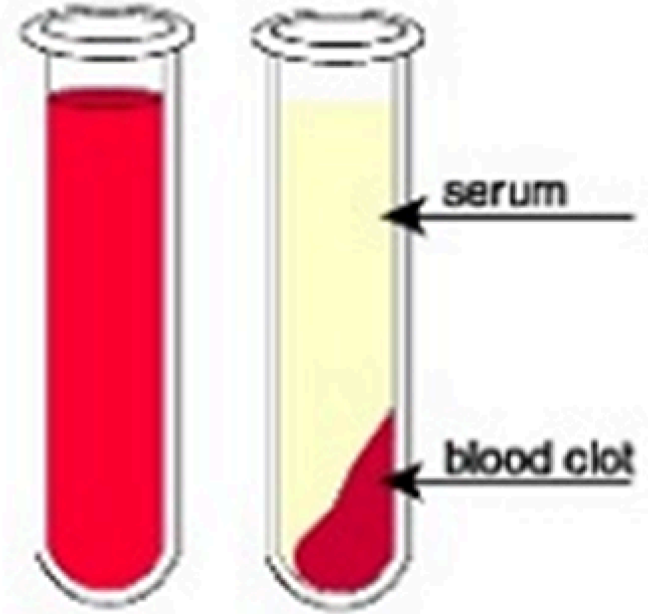
Serum is the liquid part of blood **AFTER coagulation**, therfore devoid of clotting factors as fibrinogen.

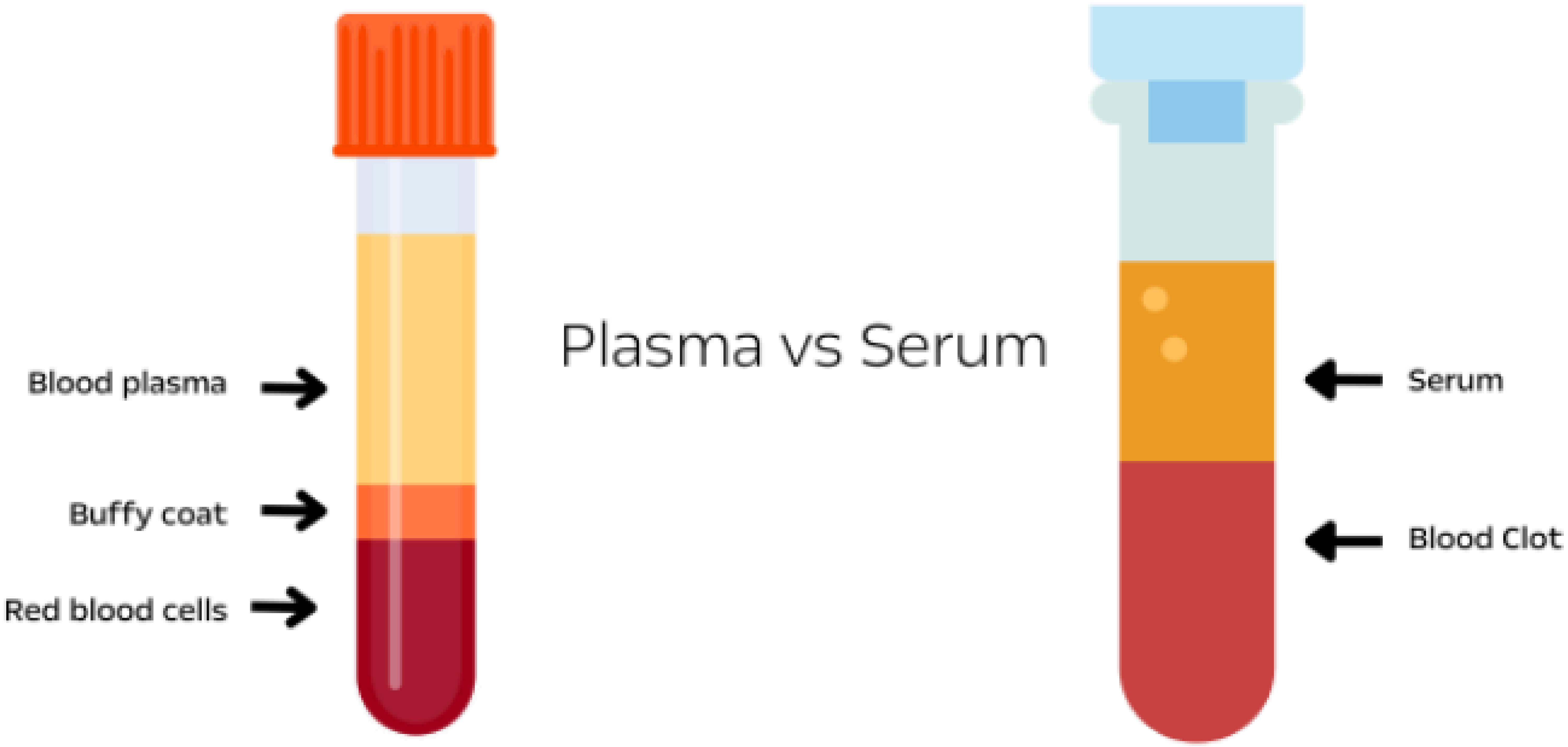
Anticoagulated

Clotted



•serum= plasma - fibrinogen

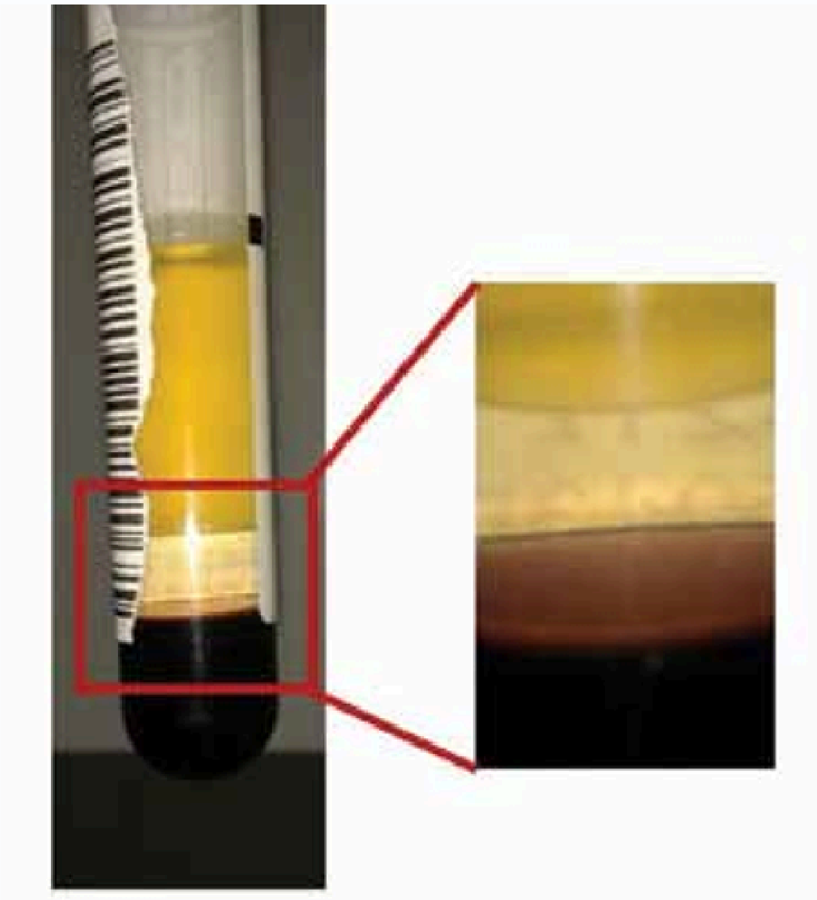



Plasma vs Serum
Blood plasma
Buffy coat
Red blood cells
Serum
Blood Clot

Serum Tubes

Serum (clot activator) tubes

- These tubes have silica particles, which activate clotting.
- Some also have a gel to separate the serum. Those without the separating gel are potentially more useful in sensitive diagnostic testing.



Plain Tube

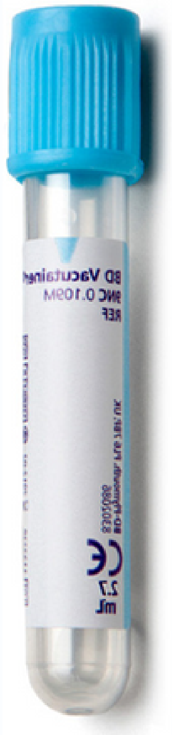
Plasma Tubes (Anticoagulant Tubes)

Plasma (Anticoagulant) Tubes

- **These tubes are coated with anticoagulant factors which prevent the clotting**

EDTA tubes

Sodium Fluoride tubes



Sodium Citrate tubes

Lithium/Sodium Heparin tubes

Bleeding of mice

Bleeding of mice is classified based on the needed amount of blood:

High blood volume

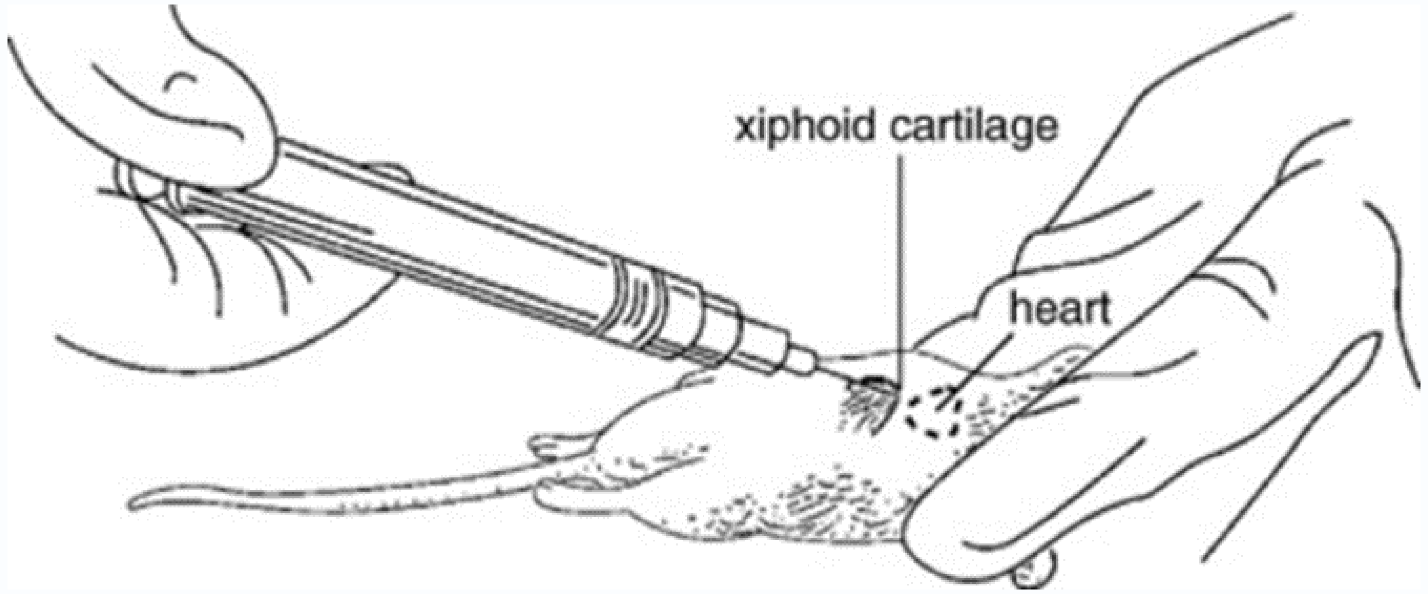
- **Decapitation (Remove of head): do not need the experimental animals anymore**

Low blood volume (The animal still alive)

- **Cardiac puncture**
- **Venous puncture**
- **Retroorbital puncture**

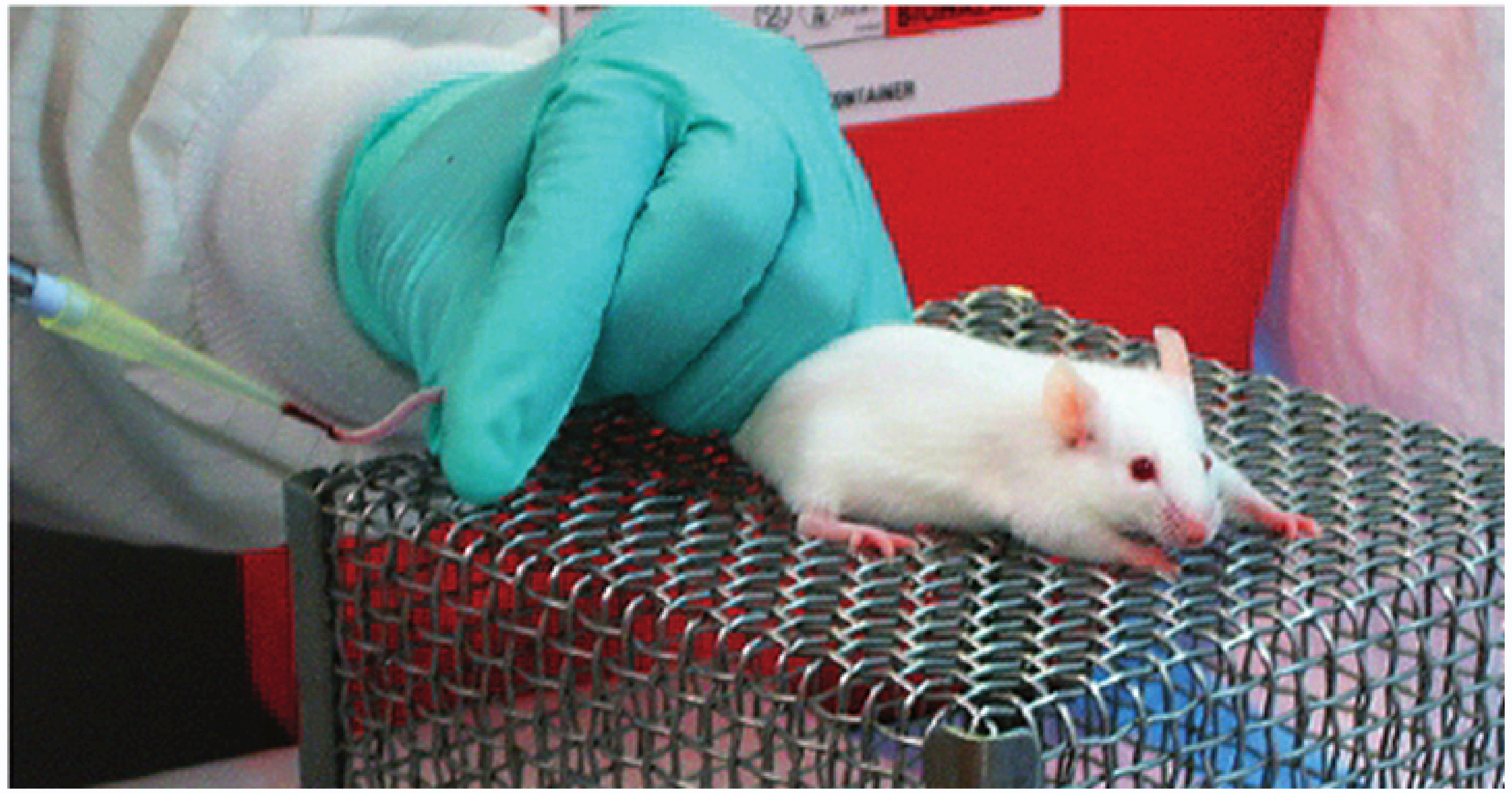
Cardiac puncture - mice

- The mouse is anesthetized and blood is collected via the left ventricle or right ventricle.
- Blood will be withdrawn slowly to prevent the heart from collapsing.
- High risk of cardiac failure
- Large amounts (~500-1000 µL, up to total blood volume).



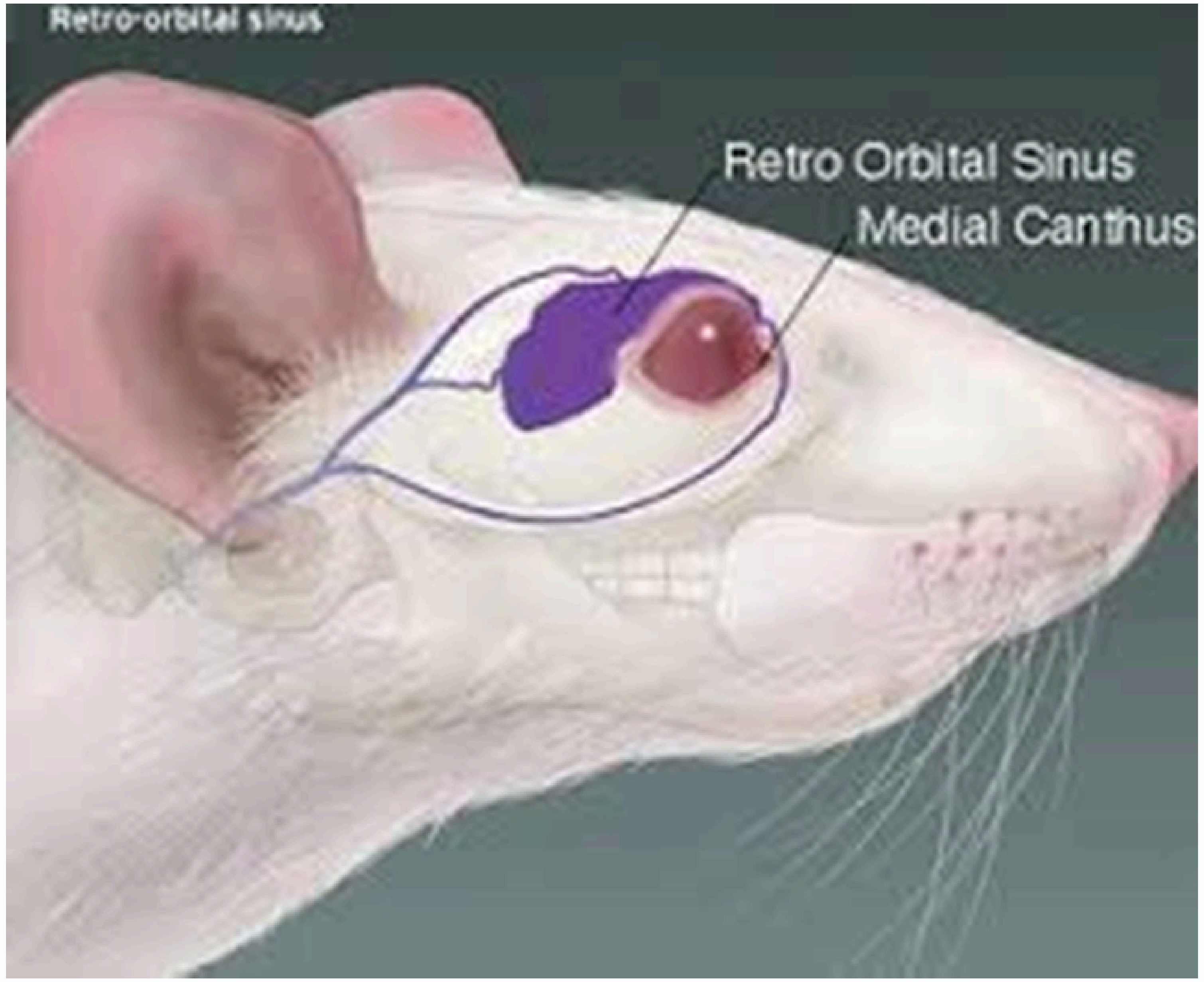
Venous puncture - mice

- From vein
- For mice: from the tail
- For rabbits: from the ear
- Xylene could be used for vasodilation
- Small amounts (~10-200 µL).



Retro orbital puncture - mice

- From Retro-orbital venous sinus (behind the eye)
- Risk of blindness
- Moderate amounts (~100-500 µL).



Serum Preparation

The serum is prepared in five steps:

1. Collecting Blood Samples
2. Coagulation
3. Separation
4. Clarification
5. Preservation.

- It is essential to use very clean glassware and plasticware to prepare and store sera.
- Sterile equipment is preferable but not essential.

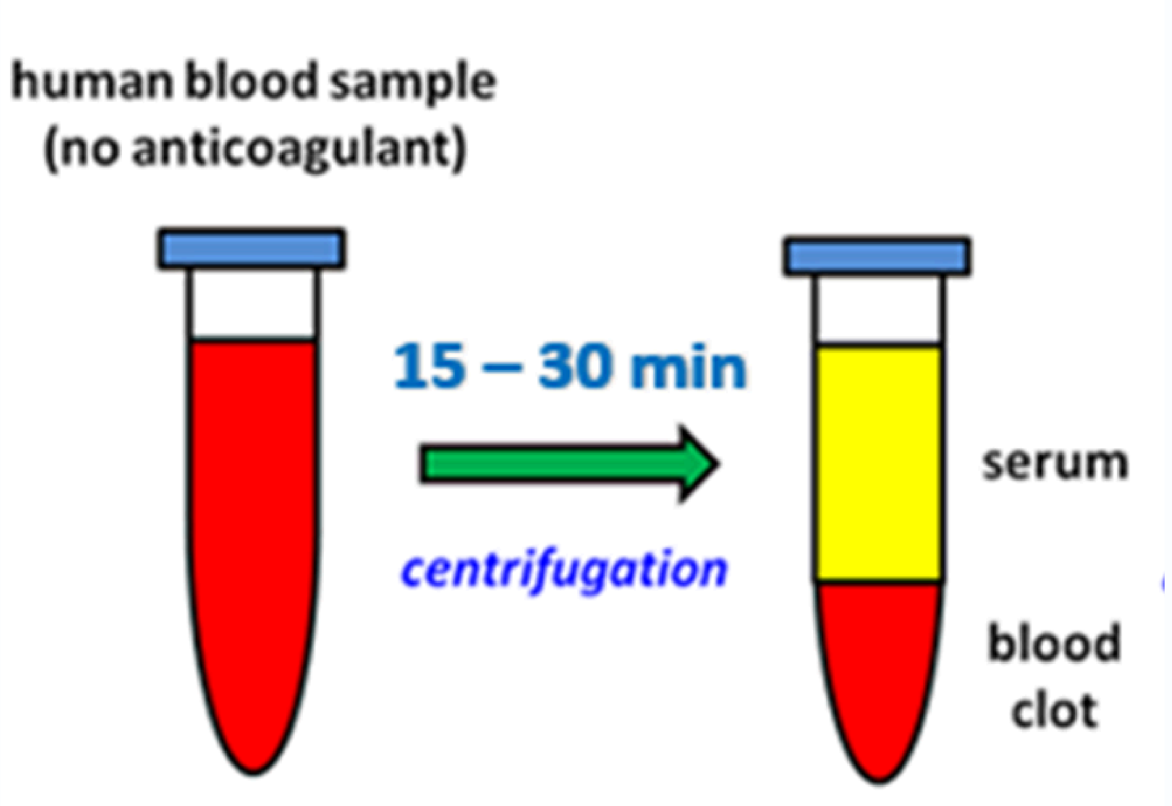
Collecting Blood Samples and Coagulation

1. Collect whole blood in Serum (clot activator/- Plain tubes) tubes.
2. After collection, the blood samples are allowed to clot by leaving them undisturbed at room temperature.
3. This usually takes about 30 minutes to an hour, depending on the sample volume and other factors. During this time, clotting factors in the blood cause it to coagulate, forming a clot.
4. Avoid shaking or vigorous handling to prevent hemolysis.

- Avoid anticoagulants like EDTA or heparin, as they prevent clotting and yield plasma instead of serum.

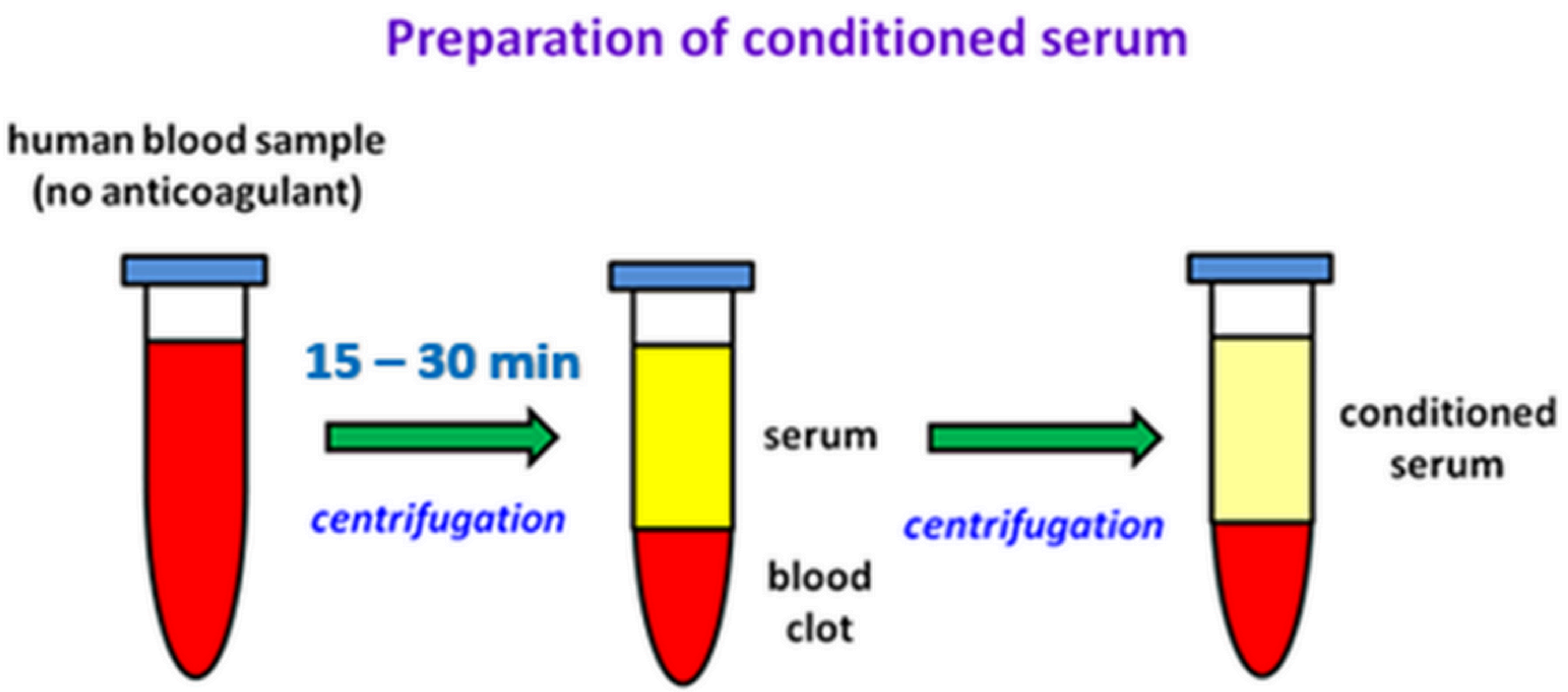
Separation

1. Once the blood has clotted, it is centrifuged to separate the serum from the clot.
2. Centrifugation involves spinning the blood samples at high speeds (2,500 rpm
3. for 10 minutes), which causes the heavier components (such as the clot) to settle at the bottom of the tube, while the lighter components (including the serum) remain suspended above it.
4. Following centrifugation, it is important to immediately transfer the liquid component (serum) into an Eppendorf tube using a Pasteur pipette



Clarification

1. Turbidity of serum is a sign of high lipid content which must be clarified as lipids interfere with the visibility of antigen-antibody reaction in vitro
2. High lipid content in the serum is not desirable as it may interfere with serological analyses and can cause embolism in animals when the serum is injected
3. To reduce the quantity of lipids in serum:
    - By Ultracentrifuge (250000 rpm for 15 min)
    - The animals should be fasted for 12 hours prior to bleeding.



Preservation and storage

1. To store the serum for long periods, add a preservative to kill microorganisms or to arrest their growth by adding one of the following serum preservatives at the following final concentrations.
    - Merthiolate
    - Sodium azide
    - Phenol

2. To preserve IgG and IgM antibodies store the serum at - 20C or below.
3. To preserve complement activity and IgE antibodies store the serum at - 80C or below.
4. Lyophilized serum powder can be stored at 4C (Cryopreservative)

Plasma Preparation

1. Collect whole blood into commercially available anticoagulant-treated tubes
2. Cells are removed from plasma by centrifugation for 10 minutes at 1,000–2,000 rpm using a refrigerated centrifuge.
3. Centrifugation for 15 minutes at 2,000 rpm depletes platelets in the plasma sample.
4. The samples should be maintained at 2–8°C while handling.
5. If the plasma is not analyzed immediately, the plasma should be apportioned into 0.5 ml aliquots, stored, and transported at –20°C or lower.

The End